Lecture 17

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Recap

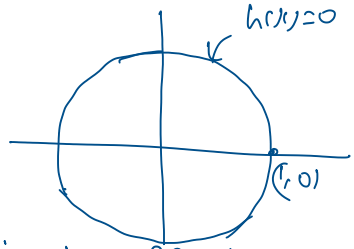
$$\min f(x)$$
$$\text{s.t. } \underbrace{h_i(x)=0,\ i=1,\dots,m}_{h(x)=0} \qquad h:\mathbb{R}^n \to \mathbb{R}^m$$

Feasible direction fix. x s.t. $h(x)=0$

d is called a feasible direction at x if there exists an $\epsilon>0$ s.t.

$$h(x+\alpha d)=0 \qquad \forall\, \alpha\in(0,\epsilon)$$

Example $h(x) = x_1^2+x_2^2-1$



there is no feasible direction at $(1,0)$

First order feasible direction

d is called a first order feasible direction at x if

$$h(x+\alpha d) = o(\alpha) \checkmark$$

Note $f(\alpha)=o(\alpha) \iff \lim_{\alpha\to 0}\frac{f(\alpha)}{\alpha}=0$

$$\Rightarrow \lim_{\alpha\to 0}\frac{h_i(x+\alpha d)}{\alpha}=0 \qquad \forall\, i$$

$$\Rightarrow \lim_{\alpha\to 0}\frac{h_i(x+\alpha d)-h_i(x)}{\alpha}=0$$

$$\Rightarrow \nabla h_i(x)^T d = 0 \quad \forall\, i=1,\dots,m$$

Define a matrix

$$\nabla h(x) = \begin{bmatrix} | & & | \\ \nabla h_1(x) & \cdots & \nabla h_m(x) \\ | & & | \end{bmatrix}_{n\times m}$$

$$\nabla h(x)^T = \begin{bmatrix} \text{---} & \nabla h_1(x)^T & \text{---} \\ \text{---} & \nabla h_m^T(x) & \text{---} \end{bmatrix}_{m\times n}$$

$$\Rightarrow \nabla h(x)^T d = 0$$

Connection b/w feasible directions and first order feasible directions

Suppose d is a feasible direction at x (x is feasible $\iff h(x)=0$)

$\forall\, \alpha$ $\quad h_i(x+\alpha d)=0 \quad \forall\, \alpha\in(0,\epsilon)$

$$\Rightarrow \underbrace{h_i(x)}_{=0} + \alpha \nabla h_i(x)^T d + o(\alpha) = 0 \quad \forall\, \alpha\in(0,\epsilon)$$

$$\Rightarrow \alpha\, \nabla h_i(x)^T d + o(\alpha) = 0 \quad \forall\, \alpha\in(0,\epsilon)$$

$$\Rightarrow \nabla h_i(x)^T d + \frac{o(\alpha)}{\alpha} = 0 \quad \forall \alpha \in (0, \bar{\alpha})$$

$$\Rightarrow \lim_{\alpha \to 0} \left(\underbrace{\nabla h_i(x)^T d + \frac{o(\alpha)}{\alpha}} \right) = 0$$

$$\Rightarrow \nabla h_i(x)^T d = 0$$

we can argue it for all $i = 1, \dots m$

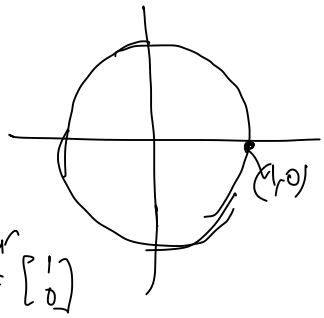
$$\Rightarrow \nabla h(x)^T d = 0$$

i.e. d is first order feasible

A first order feasible direction need not be a feasible direction

Example

$$h(x) = x_1^2 + x_2^2 - 1$$



first order feasible directions at $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\nabla h(x)\Big|_{x=\begin{bmatrix}1\\0\end{bmatrix}} = \begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix}_{x=\begin{bmatrix}1\\0\end{bmatrix}} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$[2, 0]\, d = 0 \Rightarrow 2d_1 + 0 \times d_2 = 0$$

$$\Rightarrow d_1 = 0$$

So any $d = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$ s.t. $d_1 = 0$ is a first order feasible direction.

$A^T d = 0$
$\Rightarrow d$ is in null space of A^T

if d is first order feasible dir^n at x then it is in null space of $\nabla h(x)^T$

$$V(x) = \{ \underbrace{d \in \mathbb{R}^n : \nabla h(x)^T d = 0}_{\text{null space of } \nabla h(x)^T} \}$$

set of all first order feasible directions at x.

Regular point

x is called a regular point if

① $h(x) = 0$ $\quad$ x is feasible

and ② $\underbrace{\nabla h_1(x)}_{\in \mathbb{R}^n}, \dots \nabla h_m(x)$ are linearly independent

First order feasible directions: Given a $x \in \mathbb{R}^n$, $d \in \mathbb{R}^n$ is called a first order feasible direction at x if $\nabla h(x)^T d = 0$ ($\nabla h_i(x)^T d = 0, i = 1, \cdots, m$).

- Each feasible direction is a first order feasible direction but each first order feasible direction need not be feasible.

Set of first order feasible directions at x

$$\cancel{V(x)} = \left\{d \in \mathbb{R}^n : \nabla h(x)^T d = 0\right\}$$

- This null space of $\nabla h(x)^T$.

Regular points: A $x \in \mathbb{R}^n$ is called a regular point if

1. $h(x) = 0$ (x is feasible)
2. $\nabla h_i(x), i = 1, \cdots, m$ are linearly independent.

Recall the optimization problem

$$\min f(x) \quad \text{s.t.} \quad h(x) = 0 \qquad \begin{pmatrix} h_1(x) = 0 \\ h_2(x) = 0 \\ h_m(x) = 0 \end{pmatrix}$$

Lagrange Multiplier Theorem (necessary conditions for local optimality): Let f and h be continuously differentiable. Let x^* be a local minimum that is also regular. Then there exists a unique [Equation], called Lagrange multiplier, such that

[Equation]

If in addition [Equation] and [Equation] are twice continuously differentiable then

[Equation] for all [Equation].

- These are also sufficient conditions if [Equation] is convex and the constraint set [Equation] is also convex.

$$\nabla f(x) + \nabla h(x) \lambda^* = 0$$

$$\Leftrightarrow \nabla f(x) + \begin{bmatrix} \nabla h_1(x) & \cdots & \nabla h_m(x) \end{bmatrix} \begin{bmatrix} \lambda_1^* \\ \vdots \\ \lambda_m^* \end{bmatrix} = 0$$

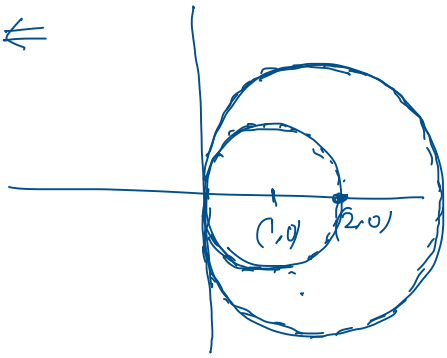
$$\Leftrightarrow \nabla f(x) + \sum_{i=1}^{m} \lambda_i^* \nabla h_i(x) = 0$$

$$d^T \underbrace{\nabla^2 f(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla^2 h_i(x^*)}\, d \geq 0 \quad \forall d \in V(x^*)$$

- the condition may be satisfied even if this matrix is not positive semi definite

Example

$$\min x_1 + x_2$$
$$\text{s.t.} \quad (x_1 - 1)^2 + x_2^2 - 1 = 0 \Leftarrow$$
$$(x_1 - 2)^2 + x_2^2 - 4 = 0$$



only feasible point is $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$x^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

necessary conditions

$$\nabla f(x) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\nabla h(x) = \begin{bmatrix} 2(x_1 - 1) & 2(x_1 - 2) \\ 2x_2 & 2x_2 \end{bmatrix}$$

$$\nabla h(x) \Big|_{x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}} = \begin{bmatrix} -2 & -4 \\ 0 & 0 \end{bmatrix}$$

there must be $\lambda^* = \begin{bmatrix} \lambda_1^* \\ \lambda_2^* \end{bmatrix}$ s.t.

$$\nabla f(x^*) + \lambda_1^* \nabla h_1(x^*) + \lambda_2^* \nabla h_2(x^*) = 0$$

$$\nabla f(x^*) + \lambda_1^* \nabla h_1(x^*) + \lambda_2^* \nabla h_2(x^*) = 0$$

$$\begin{bmatrix}1\\1\end{bmatrix} + \lambda_1^* \begin{bmatrix}-2\\6\end{bmatrix} + \lambda_2^* \begin{bmatrix}-4\\0\end{bmatrix} = 0$$

$$\Rightarrow 2\lambda_1^* + 4\lambda_2^* = 1$$

$$0\lambda_1^* + 0\lambda_2^* + 1 = 0$$ — can not be satisfied

$\nabla h_1(x^*)$ and $\nabla h_2(x^*)$ are not linearly independent

$\Rightarrow x^*$ is not a regular point

$\Rightarrow$ Hence the two conditions of the above theorem are not necessary in this case

$x^* \in \mathbb{R}^n$, $\lambda^* \in \mathbb{R}^m$ (m+n unknowns)

Approach to obtain [Equation] : (a) Solve

[Equation] (m equations)

[Equation] — (n equations)

for [Equation] unknowns [Equation] and [Equation] .

(b) Impose second order necessary conditions if applicable.

(c) Inquire each of the candidate solutions.

some will pass

some will fail

- Compute $f(x^*)$ for each x^*

$\Rightarrow$ look for x^* that gives the minimum value.

Example

$$\min \frac{1}{2}(x_1^2 + x_2^2 + x_3^2)$$

$$\text{s.t. } x_1 + x_2 + x_3 = 3 \quad \text{—— (1)}$$

$$h(x) = x_1 + x_2 + x_3 = 3$$

$$\nabla f(x) + \nabla h(x) \cdot \lambda = 0$$

$$\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} + \lambda \begin{bmatrix}1\\1\\1\end{bmatrix} = 0 \quad \text{—— (2)}$$

From (2)

$$x_1 = x_2 = x_3 = -\lambda$$

From (1) $-3\lambda = 3 \Rightarrow \lambda = -1$

$$x^* = \begin{bmatrix}1\\1\\1\end{bmatrix}, \quad \lambda^* = -1$$ is the unique soln.

$$\nabla h(x^*) = \begin{bmatrix}1\\1\\1\end{bmatrix}$$ (trivially linearly independent)

$\Rightarrow x^*$ is regular.

2nd order cond

$$\nabla^2 f(x) = \begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix} = I$$

$$\nabla^2 h(x) = \begin{bmatrix}0\end{bmatrix}$$

$$d^T(\underbrace{I + \lambda[0]}_{PD})\, d \geq 0 \qquad \forall\, d \in V(x^*)$$

this condition trivially holds

So, 2nd order necessary conditions are also satisfied.

But the objective fn is a convex fn and the constraint set also is a convex set

Hence the necessary conditions are also sufficient

So, $x^* = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix}$ is a global minimum (unique)

Define $L : \mathbb{R}^{n \times m} \longrightarrow \mathbb{R}$

$$L(x, \lambda) = f(x) + \lambda^T h(x) = f(x) + \sum_{i=1}^{m} \lambda_i h_i(x)$$

$$\nabla_x L(x, \lambda) = \nabla f(x) + \sum_{i=1}^{m} \lambda_i \nabla h_i(x)$$

$$\nabla^2_{xx} L(x, \lambda) = \nabla^2 f(x) + \sum_{i=1}^{m} \lambda_i \nabla^2 h_i(x)$$

Lagrange function: [Equation]

[Equation]

Necessary conditions: Let [Equation] be a local minimum that is also regular. Then there exists a unique [Equation]

[Equation]

- [Equation] for all [Equation]

Sufficient conditions for local optimality: Let [Equation] and [Equation] are twice continuously differentiable and [Equation] and [Equation] be such that

[Equation]

[Equation]

- [Equation] for all [Equation]

Then [Equation] is a strict local minimum.

Example

$$\min \; -(x_1 x_2 + x_2 x_3 + x_3 x_1)$$

$$\text{s.t.} \quad x_1 + x_2 + x_3 - u = 0$$

$$h(x) = x_1 + x_2 + x_3 - u \quad \text{—— (1)}$$

First order condition

$$\nabla f(x) + \nabla h(x)\, \lambda = 0$$

$$\begin{bmatrix} -x_2 - x_3 \\ -x_1 - x_3 \\ -x_1 - x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \lambda = 0 \quad \text{—— (2)}$$

unique solution to (1) and (2)

$$x^* = \begin{bmatrix} u/3 \\ u/3 \\ u/3 \end{bmatrix} \qquad \lambda^* = 2u/3$$

2nd order conditions

$$\nabla^2_{xx} L(x, \lambda) = \nabla^2 f(x) + \lambda\, \nabla^2 h(x)$$

$$\nabla^2_{xx} L(x^*, \lambda^*) = \begin{bmatrix} 0 & -1 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & 0 \end{bmatrix}$$ (this is not positive definite)

We only need $d^T \begin{bmatrix} 0 & -1 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & 0 \end{bmatrix} d > 0 \quad \forall\, d \in V(x^*) \setminus \{0\}$

$[1\ 1\ 1] \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$

$V(x^*) = \{d : \nabla h(x^*)^T d = 0\}$

$= \{d : d_1 + d_2 + d_3 = 0\}$ ✓

For any d s.t. $d_1 + d_2 + d_3 = 0$, $d \neq 0$

$$[d_1\ d_2\ d_3] \begin{bmatrix} 0 & -1 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & 0 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} = [d_1\ d_2\ d_3] \begin{bmatrix} -d_2 - d_3 \\ -d_1 - d_3 \\ -d_1 - d_2 \end{bmatrix}$$

$$= [d_1\ d_2\ d_3] \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} = d_1^2 + d_2^2 + d_3^2 > 0$$

Hence sufficient conditions are satisfied.

So, the unique optimal solution is

$$x^* = \begin{bmatrix} -u/3 \\ -u/3 \\ -u/3 \end{bmatrix}$$

optimal value $p(u) \triangleq f(x^*) = -3 \times \frac{u^2}{9} = -\frac{u^2}{3}$

$\Rightarrow p'(u) = \frac{-2u}{3}$

$= -\lambda^*(u)$

$\min f(x)$
s.t. $h(x) = u$, $u \in \mathbb{R}^m$

opt soln $x^*(u)$

Lagrange multiplier $\lambda^*(u)$

✓ opt value $p(u) = f(x^*(u))$

$\nabla p(u) = -\lambda^*(u)$

Sensitivity: Let [Equation] be an optimal solution and [Equation] be the optimal value. The rate of change of the optimal value with respect to [Equation], [Equation], is referred to as sensitivity.

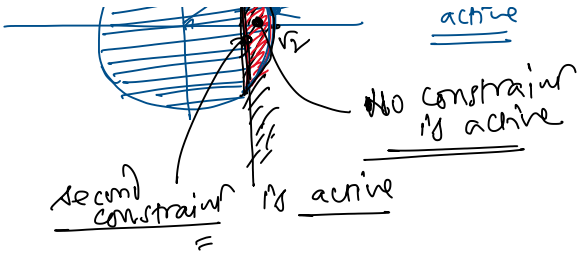
① $h(x^*(u)) = 0$
② $\nabla h_1(x^*(u)) \ldots \nabla h_m(x^*(u))$ are linearly ind.

Sensitivity Theorem: Let [Equation] be regular and [Equation] be an optimal solution – Lagrange multiplier pair that satisfies the sufficient conditions then

[Equation]

$\min f(x)$
s.t. $x_1^2 + x_2^2 - 2 \le 0$
$1 - x_1 \le 0$ $\Big\}$ — $(x_1 \ge 1)$

first constraint is



Inequality constraints: [Equation]

[Equation]

s.t. [Equation] ([Equation])

[Equation] ([Equation])

Active inequality constraints: For a [Equation] , the set of active inequality constraints

[Equation]